

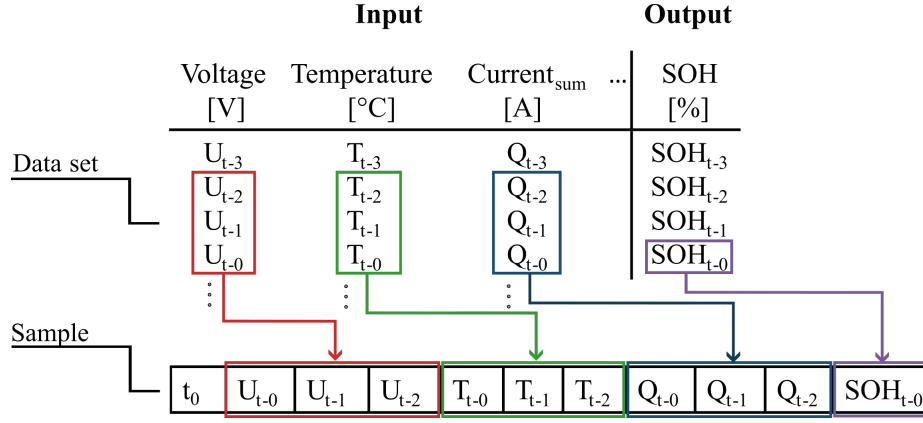


Figure 5.2: Illustration of lag-sequence construction from raw cell measurements into supervised MLP samples. Voltage, temperature, and cumulative current form the structured input history, while the SOH at the current time step serves as the corresponding label.

A configuration of $\Delta t = 30$ min and $L = 12$, corresponding to 6 h of history, is used as the initial search point. It provides a broad window relative to the approximately 20 min to 1 h charging and discharging intervals present in the investigated scenarios. The choice is subsequently evaluated rather than retained without comparison. The tested time resolutions are 1, 10, 30, and 60 min, combined with 8, 12, 16, and 20 lag steps. Among these combinations, 16 steps at a resolution of 10 min, representing 160 min of measurement history, yield the lowest error across the exemplary cells considered in Section 5.5.

The time resolution has to match the actual charging and discharging behaviour. If the spacing is too coarse, relevant dynamics disappear. If it is too fine, a fixed number of steps covers only a short operating interval, while extending the number of steps increases the input dimension and can make training less effective. If short pulses fall within one interval, they are represented only through the aggregated feature values of that interval. This design can also be linked to the sampling theorem: the spacing between data points has to remain compatible with the temporal structure of the charging and discharging cycles so that the network can still recognise the relevant periodic behaviour.[107] Because the raw measurements are resampled before sequence generation, the preprocessing step also acts like a low-pass filter and strongly attenuates high-frequency noise components, so noise is not treated as a dominant issue for the present estimation setup.

The lag sequence does not predict a future voltage, current, or temperature trajectory. Instead, the model learns a mapping between the recent operating history